

María G. Mendoza

Berkeley, CA
[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

maria_mendoza@berkeley.edu
661-714-8024

RESEARCH INTERESTS

Multi-Agent Systems, Reinforcement Learning, Distributed Optimization, Game Theory, Mechanism Design, Learning-Enabled Control, Sequential Decision Making under Uncertainty, Human–Autonomy Interaction, Cyber-Physical Systems, Humanitarian and Societal-Scale Systems.

EDUCATION

University of California, Berkeley – Ph.D. Mechanical Engineering (3.9/4.0) 2023 - Present
Advisor: Prof. Shankar Sastry
Research Area: Multi-agent systems design and control

Purdue University – M.S. Aeronautics and Astronautics 2021
University of Southern California – B.S. Mechanical Engineering 2017

PUBLICATIONS

JOURNALS

Mendoza, M. G., Maheshwari, C., Tuck, V. M., Su, P. Y., Qin, V. L., Seshia, S. A., ... & Sastry, S. “Privacy-Preserving Mechanisms for Coordinating Airspace Usage in Advanced Air Mobility (AAM)”. **ACM Journal of Autonomous Transportation Systems**, 2025.

- Developed a market-based mechanism for privacy-preserving airspace allocation using distributed optimization, Fisher markets, and competitive equilibrium pricing, improving airspace utilization, reducing delays, and increasing social welfare.

CONFERENCE PAPERS

M.G. Mendoza, A. Kalanther, D. Bostwick, E. Stephan, C. Maheshwari, and Shankar Sastry. “Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments.” **IEEE Global Humanitarian Technology Conference (GHTC)**, 2025.

- Developed a Partially Observable Markov Decision Process (POMDP)-based multi-UAV coordination framework using recurrent Proximal Policy Optimization (PPO) and psychology-informed agent-based human behavior models, reducing evacuation time and improving system resilience under uncertainty.

Choi, J., Aloor, J., Li, J., **Mendoza, M.G.**, Tomlin, C., Balakrishnan, H. “Resolving Conflicting Constraints in Multi-Agent Reinforcement Learning with Layered Safety”. **Robotics Science and Systems (RSS) Conference**, 2025.

- Proposed a layered safety framework combining Control Barrier Functions with multi-agent reinforcement learning to enforce safety constraints in robotic coordination, demonstrating robust performance in hardware experiments and Advanced Air Mobility simulations.

WORKSHOP PAPERS

MG Mendoza, L Waldburger, J Lee, Shankar Sastry. “*Simulating Human Sequential Decision-Making During Disasters.*” Accepted to **ICLR 2026 Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI** (Poster).

- Developed a persona-driven framework for modeling sequential human decision-making in disasters using hierarchical planning and Large Language Model (LLM)-based reasoning.

M.G. Mendoza, Tuck Victoria, C. Maheshwari, Shankar Sastry “*Decentralized Ergodic Coverage Control in Unknown Time-Varying Environments.*” Accepted to **AAMAS 2026 Autonomous Robots and Multirobot Systems (ARMS) Workshop** (Oral).

- Developed a decentralized multi-agent coverage algorithm using Gaussian Process belief updates and ergodic Markov chain policies for exploration in dynamic environments.

OTHER

Lockheed Martin Internal Publications (classified, 2017 - 2022)

“*Reduction of Semi-Truck Aerodynamic Drag.*” Presented at the **American Institute of Aeronautics and Astronautics (AIAA)** Region VI Student Paper Conference, 2017

- Designed and evaluated aerodynamic modifications for semi-trucks using wind tunnel testing and computational fluid dynamics (CFD) validation, demonstrating significant drag reduction and improved fuel efficiency potential.

RESEARCH EXPERIENCE

Graduate Research Assistant, University of California, Berkeley 2023 - Present

- Develop algorithms for modeling and controlling **multi-agent systems** using **distributed control, optimization, learning-based methods, game theory, and mechanism design** in uncertain, dynamic, and complex environments.
- Research applies to **Advanced Air Mobility (AAM)**, autonomous robotics for **humanitarian assistance**, strategic vehicle path planning, dynamic routing, and safety-enabled multi-agent reinforcement learning (MARL) methods for collision avoidance.

Undergraduate Research, University of Southern California 2017

- Supervised by Prof. Yan Jin at the IMPACT Laboratory, focusing on machine learning prediction algorithms for mechanical design.
- Developed and evaluated algorithms for ship collision avoidance, gaining skills in data analytics, applied mathematics, communication, and technical writing.

Undergraduate Research, Tsinghua University, Beijing, China 2016

- Worked under Prof. Mingguo Zhao in the Robotic Controls Lab.
- Designed and tested mechanical components for passive walking in bipedal robots, studying human locomotion and passive-dynamic mechanisms.

Undergraduate Research, California State University, Northridge 2014

- Assisted with battery material selection, corrosion analysis, and potentiostatic testing in the Materials Lab, supervised by Dr. Reiner and Dr. Bavarian.

PROFESSIONAL EXPERIENCE

- Research Scientist**, Lockheed Martin 2017-2022
- Led multiple AI/ML and robotics research projects, focusing on hypersonic systems, human-drone communication, and mission exploration in constrained environments.
 - Managed research with five partner universities on hypersonic technologies and proposed novel aircraft weight prediction methods.
 - Performed aerothermal analysis to develop next-gen high-speed vehicle avionics and structure.

TEACHING EXPERIENCE

- Graduate Student Instructor**, University of California, Berkeley Spring 2026
- EE 290 - Learning Enabled Multi-Agent Systems
- Engineering Faculty**, Cañada College, San Mateo Community College District 2022-2023
- Developed and taught **Introduction to Engineering** and **Computational Methods for Engineers and Scientists** in a hybrid format.
 - Mentored and advised students on academic and career pathways in engineering.
 - Designed inclusive curriculum materials to support transfer readiness for diverse student populations.

PRESENTATIONS

- Presentation of Technical Paper “Coordinated Autonomous Drones for Human-Centered Fire Evacuation in Partially Observable Urban Environments” at IEEE Global Humanitarian Technology Conference (2025), DARPA Assured Neuro Symbolic Learning and Reasoning (ANSR) Meetings 2025-2026
- **Invited Speaker**, *Encuentro de Mujeres Latinas en STEM del Área de la Bahía (Symposium hosted by the Chilean Government)*.
Talk: “*Un renacer científico en el extranjero: Una travesía entre lo profesional, la readaptación cultural y el proceso de autoconocimiento*” (“*A Scientific Rebirth Abroad: A Journey Through Professional Growth, Cultural Readaptation, and Self-Discovery*”). September 12, 2024.
- Presentation of Technical Paper “Reduction of Semi-truck Aerodynamic Drag” at AIAA (2017)

MENTORSHIP AND OUTREACH

- Graduate Mentor**, UC Berkeley Transfer-to-Excellence (TTE) Research Program Summer 2025
- Designed and supervised independent research projects for two community college students in robotics and AI.
 - Project 1: Modeling human behavior for human–robot interaction.
 - Project 2: Simulating dynamic environments with fire modeling techniques for UAV disaster relief in Unity physics engine.
 - Guided students through the full research process: from project design to data analysis and poster preparation, culminating in poster presentations at the program’s research symposium.

Mentor, Upward Bound (College of the Canyons (COC)) Summer 2015

LEADERSHIP AND SERVICE

Treasurer, Latino Association of Graduate Students in Engineering & Science (LAGSES), UC Berkeley, 2026
Corporate Liaison, LAGSES, UC Berkeley, 2025

Outreach Chair, SHPE, University of Southern California, 2016

Chapter Founder & President, Society of Hispanic Professional Engineers (SHPE), COC, 2014

PROFESSIONAL MEMBERSHIPS

- Society of Hispanic Professional Engineers (SHPE)
- Institute of Electrical and Electronics Engineers (IEEE)
- Latino/a Association of Graduate Students in Engineering and Science (UC Berkeley)
- Women in Robotics

HONORS AND AWARDS

- NSF Research Fellowship for Digital Transformation of Development (2024)
- Extraordinary Engineering and Technology Award, Lockheed Martin (2021)
- The Tony Maxworthy Award for combined experimental and analytical excellence (USC, 2017)
- Third Place at the Student Paper Conference at AIAA for a presentation on Semi-truck Aerodynamic Drag Reduction (2017)

SPECIALIZED SKILLS

- **Control & Optimization:** Distributed Optimization, Game Theory, Mechanism Design, Classical Control Theory, Trajectory Optimization, Safety-Critical Control
- **Reinforcement Learning & Machine Learning:** Multi-Agent Reinforcement Learning, Deep RL (PPO), Probabilistic Modeling, Gaussian Processes, Supervised & Unsupervised Learning, PyTorch, TensorFlow
- **Robotics & Autonomous Systems:** Multi-Agent Systems, UAV Systems, Human-Robot Interaction, ROS, Gazebo, AirSim, Isaac Sim, Agent-Based Simulation,
- **Programming & Tools:** Python, MATLAB, C++, Linux, Docker, Git
- **CAD & Engineering Software:** Siemens NX, SolidWorks, CREO, CATIA V5, STAR CCM+, Thermal Desktop, SINDA, FEMAP
- **Other Technical Expertise:** Aerodynamics & CFD, experimental methods (wind tunnel testing)